

GLOSSARY

A

ABDOMEN The hind part of the body, lying below the rib cage in mammals, and behind the thorax in arthropods.

ADIPOSE FIN A small fin lying between the dorsal and caudal (tail) fins of some fishes.

AMPLEXUS A breeding position adopted by frogs and toads, with the male using its front legs to hold the female. Fertilization usually occurs outside the female's body.

ANTENNA (pl. ANTENNAE) A sensory feeler on the head of an arthropod. Antennae always occur in pairs, and they can be sensitive to touch, sound, heat, and also taste. Their size and shape varies according to the way in which they are used.

ANTLER A bony growth on the head of deer. Unlike horns, antlers often branch, and in most cases they are grown and shed every year, in a cycle linked to the breeding season.

ARBOREAL Living fully or partly in trees.

ARTICULATION A joint—for example, between adjacent bones.

ASEXUAL REPRODUCTION A form of reproduction that involves just one organism. Asexual reproduction is most common in invertebrates, and is used as a way of boosting numbers quickly in favorable conditions. See also *Sexual reproduction*.

ATRIUM A hollow entrance chamber.

B

BACULUM A bone found in the penis of some mammals.

BALEEN A fibrous substance used by some whales for filtering food from water. Baleen grows in the form of plates with frayed inner edges, which hang from a whale's upper jaw. The baleen plates trap food, which the whale then swallows.

BARBEL A sensory appendage attached to an animal's lips or near its mouth. Many bottom-dwelling fishes have barbels, which they use for finding food hidden on gravel or in mud.

BEAK A set of narrow, protruding jaws, usually without teeth. Beaks have evolved separately in many groups of vertebrates, including tortoises and turtles, and some whales.

BILATERAL SYMMETRY A form of symmetry in which the body consists of 2 equal halves, on either side of a midline. Most animals show this kind of symmetry.

BILL A bird's jaws. A bill is made of bone, with a hornlike outer covering.

BINOCULAR VISION Vision in which the 2 eyes face fully or partly forward, giving overlapping fields of view. This allows an animal to judge depth.

BIOME A characteristic grouping of living things, together with the setting in which they are found.

BIPEDAL Moving on 2 legs.

BLOWHOLE The nostrils of whales and their relatives, positioned on top of the head. Blowholes can be single or paired.

BREACHING Leaping out of the water (usually the sea), and landing back in it with a splash. Breaching is a characteristic form of behavior shown by many large whales.

BREEDING COLONY A large gathering of nesting birds.

BROOD PARASITE An animal—often a bird—that tricks other species into raising its young. In many cases, a young brood parasite kills all its nest-mates, so that it has sole access to all the food that its foster parents provide.

BROWSING Feeding on the leaves of trees and shrubs, rather than on grasses. See also *Grazing*.

C

CALCAREOUS Containing calcium. Calcareous structures—such as shells, exoskeletons, and bones—are formed by many animals, either for support or for protection.

CAMOUFLAGE Colors or patterns that enable an animal to merge with its background. Camouflage is very widespread in the animal kingdom—particularly among invertebrates—and is used both for protection against predators, and for concealment when approaching prey. See also *Mimicry*, *Cryptic coloration*.

CANINE TOOTH In mammals, a tooth with a single sharp point that is shaped for piercing and gripping. Canine teeth are toward the front of the jaws, and are highly developed in carnivores.

CARAPACE A hard shield on the back of an animal's body.

CARNASSIAL TOOTH In mammalian carnivores, a bladelike cheek tooth that has evolved for slicing through flesh.

CARNIVORE Any animal that eats meat. The word carnivore can also be used in a more restricted sense, to mean mammals in the order Carnivora.

CARRION The remains of dead animals.

CARTILAGE A rubbery substance that forms part of vertebrate skeletons. In most vertebrates, cartilage lines the joints, but in cartilaginous fishes—for example, sharks—it forms the whole of the skeleton.

CASQUE A bony growth on the head of an animal.

CAUDAL Relating to an animal's tail.

CECUM A blind-ended part of the digestive tract, which is often used in the digestion of plant food.

CELLULOSE A complex carbohydrate found in plants. Cellulose is used by plants as a building material, and it has a resilient chemical structure that animals find hard to break down. Plant-eating animals, such as ruminants, digest it with the help of microorganisms.

CEPHALOTHORAX In some arthropods, a part of the body that combines the head and the thorax. Animals that have a cephalothorax include crustaceans and arachnids.

CERCUS (pl. CERCI) A paired, hairlike appendage that is attached to the end of the abdomen in some invertebrates, particularly insects.

CHAETA (pl. CHAETAE) A short, stiff hair or bristle, found in many invertebrates. Chaetae sometimes have a sensory function, but in annelid worms, they are more often used in movement.

CHEEK TOOTH see *Molar tooth*, *Premolar tooth*.

CHELICERA (pl. CHELICERAE) In arachnids, each one of the first pair of appendages, at the front of the body. Chelicerae often end in pincers, and in spiders they are able to inject venom. In mites, they are sharply pointed, and are used for piercing food.

CHORDATE An animal belonging to the phylum Chordata, which includes all vertebrates. A key feature of chordates is their notochord, which runs the length of their bodies; it reinforces the body, yet allows it to move by bending.

CHRYsalis A hard and often shiny case that protects an insect pupa. Chrysalises are often found attached to plants, or buried close to the surface of the soil.

CLASPERS see *Pelvic spurs*.

CLASS A level used in classification. In the sequence of classification levels, a class forms part of a phylum, and is subdivided into one or more orders.

CLOACA An opening toward the rear of the body that is shared by several body systems. In some vertebrates—such as bony fishes and amphibians—the gut, kidneys, and reproductive systems all use this single opening.

CLOVEN-HOOFED Having hooves that look as if they are split in 2. Most cloven-hoofed mammals, such as deer and antelope, actually have 2 hooves, arranged on either side of a line that divides the foot in 2.

COCOON A case made of open, woven silk. Many insects spin a cocoon before they begin pupation, and many spiders spin one to hold their eggs. See also *Pupa*.

COLD-BLOODED see *Ectothermic*.

COLONY A group of animals, belonging to the same species, that spend their lives together, often dividing up the tasks involved in survival. In some colonial species, particularly aquatic invertebrates, the colony members are permanently fastened together. In others, such as ants, bees, and wasps, members forage for food independently, but live in the same nest.

COMMENSALISM A partnership between 2 organisms of different species, in which one benefits, but the other neither gains nor loses.

COMPOUND EYE An eye that is divided into separate compartments, each with its own set of lenses. Compound eyes

are a common feature of arthropods. The number of compartments they contain varies from a few dozen to thousands.

CORNICLE In aphids, one of the pair of narrow tubes at the end of the abdomen. The cornicles secrete wax that helps protect aphids from some predators and parasites.

COSTAL Relating to an animal's ribs.

COVERT A feather that covers the base of a bird's flight feather.

CREMASTER A collection of minute hooks on the hind end of a butterfly or moth pupa. The pupa hangs by these hooks from a solid support.

CRYPTIC COLORATION Coloration and markings that make an animal difficult to see against its background.

D

DELAYED IMPLANTATION In mammals, a delay between fertilization of an egg and the subsequent development of an embryo, which allows birth to occur when conditions are favorable for raising young.

DEPOSIT FEEDER An animal that feeds on small specks of organic matter that have drifted down through water, and settled on the bottom.

DETRITIVORE An animal that eats dead plant or animal matter.

DEWLAP A fold of loose skin hanging from an animal's throat.

DIASTEMA A wide gap separating rows of teeth. In plant-eating mammals, the diastema separates the biting teeth, at the front of the jaw, from the chewing teeth at the rear. In many rodents, the cheeks can be folded into the diastema, to shut off the back of the mouth while the animal is gnawing.

DIGIT A finger or toe.

DIGITIGRADE A gait in which only the fingers or toes touch the ground. See also *Plantigrade*, *Unguligrade*.

DIURNAL Active during the day.

DOMESTICATED Relating to an animal that lives fully or partly under human control. Some domesticated animals look identical to their wild counterparts, but many have been bred to produce artificial varieties (breeds), which are not found in nature. See also *Feral*.

DORSAL On or near an animal's back.

DREY The nest of a squirrel, which is built in a tree.

E

ECHOLOCATION A method of sensing nearby objects by using pulses of high-frequency sound. Echoes bounce back from obstacles and other animals, allowing the sender to build up a "picture" of its surroundings. Echolocation is used by several groups of animals, including